

LeetCode (787): Cheapest Flights With K stops.

Experiment 7: (Algorithm) with Complexity Analysis

Algorithm

START:

Step 1: Create an adjacency list from the flight data array (graph).

Step 2: Initialize a cost array with ∞ ~~for~~ for all nodes.

Step 3: Set $\text{cost}[\text{src}] = 0$

Step 4: Use a queue to store (stops, node, current cost)

Step 5: Start BFS traversal:

- if stops exceed k, skip.

- For each neighbour:

 - if cost is smaller $\rightarrow$ update and push into queue.

Step 6: At the end:

- if $\text{cost}[\text{dst}] = \infty \rightarrow$ return -1

- else $\rightarrow$ return $\text{cost}[\text{dst}]$.

TOP: